LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Practice Pack: Structure of the Atom

Chapter / Topic: Topic-based: use with whichever Class 9 book your students follow

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. Level 1 checks the ideas, Level 2 is board-style questions, and Level 3 is Foundation-level challenge.
2. Shell capacities: K = 2, L = 8, M = 8 (for the first 18 elements). Mass number = protons + neutrons.

LEVEL 1: CONCEPTS

Q1. Name the three sub-atomic particles and give the charge on each.

Q2. What did Rutherford conclude from the observation that most alpha particles passed straight through the gold foil?

Q3. Define atomic number and mass number.

Q4. What are isotopes? Give one example.

Q5. What are isobars? Give one example.

LEVEL 2: BOARD-STYLE QUESTIONS

Q6. Write the electronic configuration and valency of: (a) sodium (Z = 11), (b) oxygen (Z = 8), (c) chlorine (Z = 17), (d) sulphur (Z = 16).

Q7. Give three observations of Rutherford's alpha-particle scattering experiment and the conclusion from each.

Q8. Chlorine occurs as two isotopes: Cl-35 (75%) and Cl-37 (25%). Find the average atomic mass of chlorine.

Q9. Give two uses of isotopes.

Q10. Compare the Bohr model with the Rutherford model. What problem in Rutherford's model did Bohr solve?

LEVEL 3: CHALLENGE (FOUNDATION)

Q11. Bromine has two isotopes, Br-79 (49.7%) and Br-81 (50.3%). Calculate its average atomic mass.

Q12. The average atomic mass of an element X is 16.2 u. It has two isotopes, X-16 and X-18. Find the percentage of each.

Q13. An atom has 2 electrons in its K shell and 6 in its L shell. Identify the element, its valency, and whether it is likely to gain or lose electrons. Explain.

~ Draw the shells for each element you write about. It makes valency obvious! ~